

Requirements Engineering

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Example

- ◆ “As a customer, I want to buy PS5 from the website.”
 - This is a functional and verifiable requirement.
- ◆ Development Team’s **Expectation**: Delivering the requirement will make the customer happy.
- ◆ Sufficient? Many other concerns ...
 - [Efficiency Issue] How about the customer needs to wait 30 minutes after each click or the input to each textbox on the website to complete a purchase of an item?
 - [Scalability Issue] How about the website only allows exactly one customer to buy exactly one item at any time?
 - ...
- ◆ We must know the non-functional part of the requirements

Requirements Engineering (RE)

- ◆ RE is a process to *find out* and *structure* the **functional** and **non-functional requirements** of the software to be built.

Table 3. Classification rules [6]

No	Question	Result
1	Was this requirement stated because we need to specify...	Functional
2	... some of the system’s behavior, data, input, or reaction to input stimuli – regardless of the way how this is done?	Functional
3	... restrictions about timing, processing or reaction speed, data volume, or throughput?	Performance
4	... a specific quality that the system or a component shall have?	Specific quality
5	... any other restriction about what the system shall do, how it shall do it, or any prescribed solution or solution element?	Constraint

Example dataset [7]

Requirements Class	#Requirements	Percent	# Length
Functional	225	40.30%	39
Availability (A)	21	3.36%	19
Fast Response (F)	10	1.65%	19
Legal (L)	13	2.08%	18
Look & Feel (LF)	38	6.09%	20
Maintainability (M)	17	2.72%	28
Operational (O)	62	9.92%	20
Performance (P)	244	3.66%	22
Portability (PO)	1	0.16%	14
Scalability (SC)	21	3.36%	18
Security (SE)	66	10.50%	20
Usability (US)	47	10.72%	22
Total	625	100%	1

In many cases, a high proportion of all requirements are non-functional requirements

Many faces of non-functional requirements

- ◆ Be careful!
- ◆ Different people in different domains may use different terminologies
- ◆ Example “words” from users to mean non-functional requirements :
 - System properties/characteristics / constraints
 - Quality attributes, non-behavioral requirements, concerns, goals, extra-functional requirements, quality requirements, system attributes

Many faces of non-functional requirements

- ◆ Different application domains have different sets of major quality attributes.

Application Domain	Relevant NFRs
Banking and Finance	accuracy, confidentiality, performance, security, usability
Education	interoperability, performance, reliability, scalability, security, usability
Energy Resources	availability, performance, reliability, safety, usability
Government and Military	accuracy, confidentiality, performance, privacy, provability, reusability, security, standardizability, usability, verifiability, viability
Insurance	accuracy, confidentiality, integrity, interoperability, security, usability
Medical/Health Care	communicativeness, confidentiality, integrity, performance, privacy, reliability, safety, security, traceability, usability
Telecommunication Services	compatibility, conformance, dependability, modifiability, maintainability, performance, portability, reliability, usability
Transportation	accuracy, availability, interoperability, completeness, confidentiality, dependability, integrity, performance, safety, security, verifiability

Table 2 - The Most Commonly Considered NFRs		
NFRs	Definition	Attributes
Performance	requirements that specify the capability of software product to provide appropriate performance relative to the amount of resources needed to perform full functionality under stated conditions	response time, space, capacity, latency, throughput, computation, execution speed, transit delay, workload, resource utilization, memory usage, accuracy, efficiency, compliance, modes, delay, miss rates, data loss, concurrent transaction processing
	requirements that specify the capability of software product to operate without failure and maintains a specified level of performance when used under specified normal conditions during a given time period	completeness, accuracy, consistency, availability, integrity, correctness, maturity, fault tolerance, recoverability, reliability, compliance, failure rate/critical failure
Reliability	requirements that specify the end-user interactions with the system and the effort required to learn, operate, prepare input, and interpret the output of the system	learnability, understandability, operability, attractiveness, usability compliance, ease of use, human engineering, user friendliness, memorability, efficiency, user productivity, usefulness, likeability, user reaction time
Usability	requirements that concern about preventing unauthorized access to the system, programs, and data	confidentiality, integrity, availability, access control, authentication
Security	requirements that describe the capability of the software product to be modified that may include correcting a defect or make an improvement or change in the software	testability, understandability, modifiability, analyzability, changeability, stability, maintainability compliance
Maintainability		

Many faces of non-functional requirements

- ◆ Even in the same application domain, different types of applications have different sets of non-functional requirements

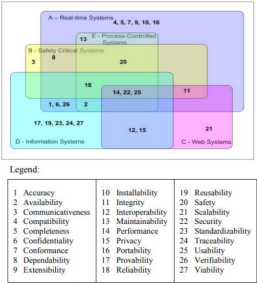


Figure 5 - Type of Systems and Relevant NFRs

Many faces of non-functional requirements

- ◆ Some quality attributes are undefined or vaguely known.
- ◆ If a quality attribute **cannot** be measured or **cannot** be verified, we **cannot** know whether we cannot predict how well our software meets them
 - For such quality attributes, we can only rely on user evaluations on the software

have definition and attributes	have definition	without definition and attributes
accessibility; adaptability; availability; efficiency; fault tolerance; functionality; integrability; integrity; maintainability; modifiability; performance; portability; privacy; readability; reliability; reusability; robustness; safety; scalability; security; testability; understandability; and usability	accuracy; analyzability; attractiveness; changeability; communicativeness; completeness; complexity; composability; confidentiality; consistency; correctness; defensibility; dependability; evolvability; extensibility; flexibility; immutability; installability; interoperability; learnability; likeability; localizability; maturity; operability; quality of service; recoverability; replicability; stability; suitability; survivability	accountability; additivity; adjustability; affordability; agility; anonymity; atomicity; availability; augmentability; certainty; compatibility; comprehensibility; comprehensiveness; conciseness; configurability; conformance; controllability; customizability; debuggability; decomposability; demonstrability; distributivity; durability; effectiveness; enhanceability; expandability; expressiveness; extensibility; feasibility; formality; generality; legibility; manageability; measurability; mobility; nomadicity; observability; predictability; provability; reconfigurability; repeatability; resticability; self-descriptiveness; simulacrum

We need to Handle Multi-Level Concerns

- ◆ specific to the application domain
 - => meet the general requirement of the industry sector
- ◆ specific to the type of software application
 - => meet the general requirement for the type of software
- ◆ the specific to the application
 - => meet the unique requirements of the current application
- ◆ In all levels, consider non-functional requirements
- ◆ If not, our application will not be used easily in the application environment

Requirements Engineering

Requirements Engineering (RE)

- ♦ **is the first step** in finding a solution for a data processing problem
- ♦ the *results* of requirements engineering is a **requirements specification**
- ♦ requirements specification is a
 - contract for the customer
 - starting point for design

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Avoid Common Mistakes in RE

[<https://ieeexplore.ieee.org/document/1695257>]

- ♦ **noise**: does not add relevant information
- ♦ **silence**: feature in the problem not mentioned in the specification
- ♦ **over-specification**: talk about the solution rather than the problem
- ♦ **contradictions**: inconsistent description
- ♦ **ambiguity**: unclear
- ♦ **forward references**: especially cumbersome in long documents
- ♦ **wishful thinking**: features that cannot realistically be realized

Natural language specs are dangerous

e.g., “All users have the same control field”

- ♦ The same value in the control field?
- ♦ The same format of the control field?
- ♦ There is one (1) control field for all users?

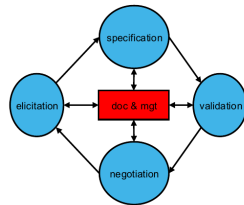
- ♦ We should *validate* the *documented* requirements to avoid misunderstanding. It leads to the RE process (see the next slide).

Requirements engineering, main steps

Four major steps in an RE process

1. understanding the problem: **elicitation**
2. describing the problem: **specification**
3. agreeing upon the nature of the problem: **validation**
4. agreeing upon the boundaries of the problem: **negotiation**

This is an iterative process



Conceptual modeling

- ◆ We **explicitly** model the scope of the requirements as a domain model
- ◆ Making a model explicit requires solving two problems:
 - **Analysis**: to find and clear ambiguity: Because of the unspoken assumptions, different languages /terminologies being spoken, incomplete codification of the problem domain
 - **Negotiation**: to resolve ambiguity due to stakeholders with different goals (e.g., workers against management).
 - As an analyst, we must participate in shaping the domain model
- ◆ Having an explicit model reduces later surprises.

How we study the world around us

- ◆ people have a set of assumptions about a topic they study
- ◆ this set of assumptions concerns/affects:
 - how knowledge is gathered
 - how the world (they know) is organized
- ◆ this results in two dimensions:
 - subjective-objective (w.r.t. knowledge)
 - conflict-order (w.r.t. the world)
- ◆ which results in 4 basic positions to requirements engineering that an analyst can take

The Positions of Analyst in RE

1. functional (objective+order): the analyst is the expert who empirically seeks the truth
2. social-relativism (subjective+order): the analyst is a 'change agent'. RE is a learning process guided by the analyst
3. radical-structuralism (objective+conflict): there is a struggle between classes; the analyst chooses for either party
4. neohumanism (subjective+conflict): the analyst is kind of a social therapist, bringing parties together

Four positions to RE

Functional

- ◆ **Functional:** functional (objective+order): the analyst is the expert who empirically seeks the truth
 - Conflicts are solved by management.
 - Management tells how things go.
 - Requirements are well-articulated, shared, objective.
 - Development is formal, rational.
 - Politics is irrelevant.
 - Reality is measurable, and the same to everyone.
 - Design is a technical problem.

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Four positions to RE

Social-Relativism

- ◆ **Social-relativism:** (subjective+order): the analyst is a 'change agent'
 - There is not one truth, there may be different perceptions.
 - An information system is part of a continually changing social environment.
 - System goals result from an organization shaping its own reality.
 - There is no objective criterion of good or bad.
 - The goal is to arrive at a consensus.

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Four positions to RE

Radical-Structuralism and Neohumanism

- ◆ **Radical-structuralism:** radical-structuralism (objective+conflict):
 - There is a struggle between classes; the analyst chooses for either party
 - There is a fundamental truth, but there also is a fundamental battle between classes.
 - You may choose for either management (machines replace people, machines guide people's work, supervision, loss of jobs, less interesting work) or for the workers (enhance labor skills, make work more interesting and challenging).
- ◆ **Neohumanism** (subjective+conflict): rather hypothetical; emancipation, remove barriers

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Requirements Elicitation

- ◆ After knowing the position we play in the RE process of a project, the next thing we do is to collect the requirements.
- ◆ The activities in the requirements elicitation process can be classified into five types.
 1. Understanding the application domain
 2. Identifying the sources of requirements
 3. Analysing the stakeholders
 4. Selecting the techniques, approaches, and tools to use
 5. Eliciting the requirements from stakeholders and other sourcesSince Elicitation is an iterative process, we cannot finish one type before starting another.

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Requirements Elicitation

1. Understanding the application domain

- ◆ Start from understanding the application domain
- ◆ The current environment of the system needs to be thoroughly explored including the political, organizational, and social aspects related to the system, in addition to any constraints they may enforce upon the system and its development.

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Requirements Elicitation

2. Identifying the sources of requirements

- ◆ Requirements spread across many sources and exist in different formats.
- ◆ People
 - Stakeholders: the main source of requirements for the system.
 - Users and subject matter experts: supply details and needs
- ◆ Things
 - E.g., Existing systems and operation processes
 - E.g., Existing documentation (e.g., manuals, forms, and reports):
 - provide useful information about the organization, environment, and rationales

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Requirements Elicitation

3. Analyzing the stakeholders

- ◆ Stakeholders are people who have an interest in the system
 - Groups and individuals internal and external to the organization.
 - Customers and direct users of the system
- ◆ Analyze and involve all the relevant stakeholders.
 - The process of analyzing the stakeholders also often includes identifying key user representatives and product champions (who support the software).
- ◆ Consult them during requirements elicitation.

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Requirements Elicitation

4. Selecting the techniques, approaches, and tools to use

- ◆ Requirements elicitation is best performed using a variety of techniques (e.g., interviewing and form analysis).
 - In many projects, several elicitation techniques are employed during and at different software development life cycle stages.

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Requirements Elicitation

5. Eliciting the requirements from stakeholders and other sources

- ◆ After the above four preparation steps, this step is the actual step to collect requirements.
- ◆ Targets
 - Establish the scope of the system
 - Investigate in detail the needs and wants of the stakeholders, especially the users
 - Determine the future processes the system will perform with respect to the business operations,
 - Examine how the system may support to satisfy the major objectives (and address the key problems) of the business

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Requirements Elicitation

Requirements Elicitation Techniques

- | | |
|---|-----------------------------------|
| ◆ interview | ◆ synthesis from existing system |
| ◆ Delphi technique | ◆ domain analysis |
| ◆ brainstorming session | ◆ Business Process Redesign (BPR) |
| ◆ task analysis | ◆ Prototyping |
| ◆ scenario analysis | ◆ mind mapping |
| ◆ ethnography | ◆ group storytelling |
| ◆ form analysis | ◆ user stories |
| ◆ analysis of natural language descriptions | ◆ Focus group |
| ◆ questionnaire | ◆ Facilitated workshop |
| | ◆ ... |

We visit the highlighted ones in the next few slides.

Requirements Elicitation

Interview

- ◆ Commonly used in requirements elicitation.
- ◆ Ask stakeholders questions about the current application's usage and the usage of the application to be built.
- ◆ Good at getting an overall understanding of stakeholders' needs and the current application's problem from their views.
- ◆ Inherently informal, and their effectiveness depends greatly on the quality of interaction between the participants.
- ◆ Interviews provide an efficient way to collect large amounts of data quickly

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Requirements Elicitation

Interview

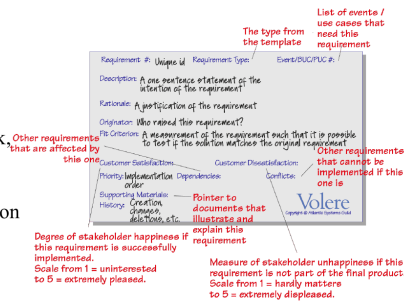
- ◆ Three types of interviews: unstructured, structured, and semi-structured (hybrid of the former two)
- ◆ Unstructured
 - Limited control over the direction of discussions.
 - Best applied for exploration when there is a limited understanding of the domain
 - Risky: Too much detail in some areas, and not enough in others

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Requirements Elicitation Interview

◆ Structured

- using a predetermined set of questions to gather specific information
- The success of structured interviews depends on knowing what are the right questions to ask, when should they be asked, and who should answer them. Templates that provide guidance on structured interviews for requirements elicitation such as Volere can be used.
- Questions can be open-ended first then close-ended (more specific).

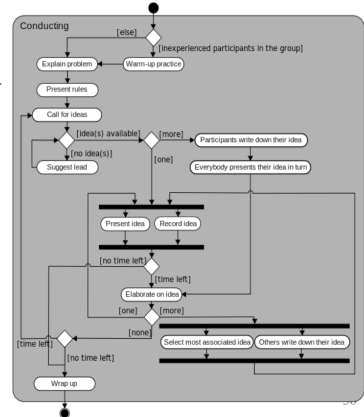


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Requirements Elicitation Brainstorming session

Osborn's method

- ◆ **Brainstorming** is a group-based activity by gathering a list of ideas iteratively toward a particular topic



Requirements Elicitation Task Analysis

- ◆ Task analysis analyzes how people perform their jobs:
 - the things they do, the things they act on, and the things they need to know.
- ◆ It employs a top-down approach where high-level tasks are decomposed into subtasks and eventually detailed sequences until all actions and events are described.
 - It determines the knowledge used or required to carry tasks out.
- ◆ E.g., how an SGS staff completes the student enrolment task, in which many typical and atypical situations need to be handled.

Requirements Elicitation Scenario-Based Analysis

- ◆ Provides a more user-oriented perspective on designing and developing an interactive system.
- ◆ E.g., observe how the SGS staff handles the enrolments of a non-local student A and a local student B. Then, ask questions on cases not yet covered.

Requirements Elicitation

Scenario-Based Analysis (example)

- ◆ first shot by observation on book return in Library.
 1. check due back date
 2. if overdue, collect fine
 3. record book as being available again
 4. put book back
- ◆ followed by a discussion with the librarian
 - what if the person returning the book is not registered as a client?
 - what if the book is damaged?
 - how to handle in case the client has other books that are overdue, and/or an outstanding reservation?

Requirements Elicitation

Form analysis

- ◆ Figure out items that are certain or have variety, and the time (past, while filling the form, or future) that the information for the item is available.

The form is titled 'Course Add/Drop Form (SCS233)' and is for the College of Business at the University of British Columbia. It includes a header with the university logo and contact information. The main body contains instructions for students, a section for student information (Name, Student No., Year of Study, Contact Phone No.), and a section for course details. The course details section has two tables: one for 'Courses to be dropped' and one for 'Courses to be added'. Each table has columns for 'Course Code', 'Course Title', 'Semester/Year', and 'Credits'. Below the tables, there are fields for the student's signature and date, and a section for the Program Director's signature and date. The form also includes a section for the student's name and student number.

Requirements Elicitation

Focus Group and Facilitated Workshop

- ◆ A **focus group** is a small, but demographically diverse, group of people whose reactions are studied in guided or open discussions.
 - people are asked about their perceptions, opinions, beliefs, and attitudes towards
 - RE Analyst records the vital points he or she is getting from the group.
- ◆ A **facilitated workshop** involves cross-functional team members to study the topic from all perspectives and make a decision as the outcome of the workshop.

Requirements Elicitation

Mind mapping, group storytelling, user stories

- ◆ Mind map

```
graph LR
    A[Software Upgrade Planning] --> B[Preparation]
    A --> C[Integration Testing]
    A --> D[Responsibilities]
    A --> E[Backup]
    B --> B1[New software license]
    B --> B2[Upgrade guide]
    B --> B3[Completeness of data]
    C --> C1[Able to save and load data]
    D --> D1[Date: Feb 20]
    D --> D2[Backup: David]
    D --> D3[Performer: Derek]
    D --> D4[Testing: Delay]
    E --> E1[Data]
    E --> E2[Software]
    E --> E3[Local configuration files]
```
- ◆ Group storytelling
 - Co-construct a story, share a story, complete an unfinished story provided by a facilitator, zoom-in/out or summarize a story, roleplay a story, analyze a story
- ◆ User story
 - E.g., As a manager, I want to browse my existing quizzes so I can recall what I have in place and figure out if I can just reuse or update an existing quiz for the position I need now

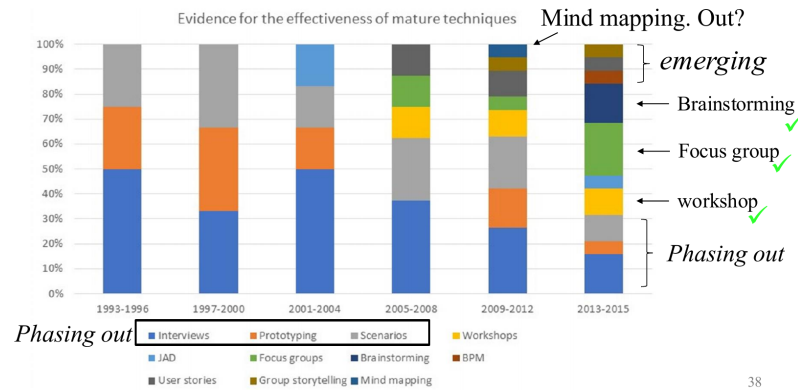
Elicitation Technique Selection: How Do Experts Do It? [4]

- ♦ **Interviewing**: particularly useful to gather initial background information when working on new projects in new domains.
- ♦ **Collaborative Sessions (focus group, workshop, brainstorming)**: Effective
- ♦ **Data Gathering from Existing Systems**: must do, but not over-analyze
- ♦ **Agile (mind mapping, group storytelling, user stories)**: popular nowadays.
- ♦ **Questionnaires**: Not effective

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Requirements Elicitation

Effectiveness of mature techniques [5]



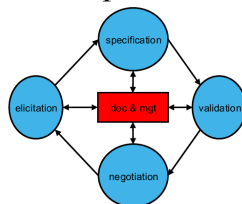
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Recall? Requirements engineering, main steps

Four major steps in an RE process

1. understanding the problem: **elicitation**
2. describing the problem: **specification**
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This is an iterative process



Specifying requirements

Structuring a set of requirements

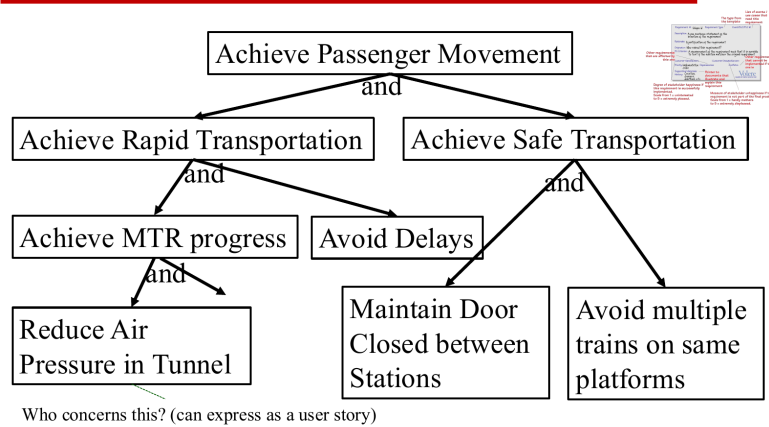
While we collect the requirements, we should structure them.

1. Hierarchical structure: Higher-level requirements are decomposed into lower-level requirements
2. Link requirements to specific stakeholders (e.g., management and end users each have their own set)

So, while we structure the requirements, we also analyze them and their relationships.

Specifying requirements

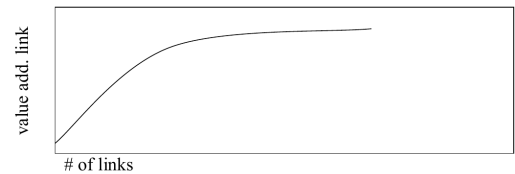
Example of Structuring Requirements



Validating Requirement

Direct versus indirect links

- ◆ Same requirements from multiple sources help us reason the validity of the requirements.
 - Lesson 1: don't rely too much on indirect links (intermediaries, surrogate users)
 - Lesson 2: The more links, the better (but up to a point)



Validating Requirement

Things to look at

- ◆ Inspection of the requirement specification w.r.t. correctness, completeness, consistency, accuracy, readability, and testability.
- ◆ some aids at different stages of development
 - Early structured walkthroughs with customers
 - Prototypes of initial versions
 - Test plan or unit testing in Agile development
 - User acceptance testing when delivery

Negotiating requirements

Prioritizing Requirements

- ◆ Apart from negotiating requirements during requirement elicitation, we should also negotiate requirements with stakeholders when prioritizing them.
- ◆ E.g., label a requirement item as “high”, “medium”, or “low” priority,
- ◆ E.g., Classify it using the MoSCoW Method
 - **Must haves**: mandatory requirements
 - **Should haves**: important but not mandatory
 - **Could haves**: if time allows
 - **Won't haves**: not today (may be tomorrow)

Example Practices in Agile Requirement Management

Practice
1. Face-to-face communication
2. Customer involvement
3. User stories
4. Iterative requirements
5. Requirements prioritisation
6. Change management
7. Cross-functional teams
8. Prototyping
9. Testing before coding
10. Requirements modelling
11. Requirements management
12. Review meetings and acceptance tests
13. Code refactoring
14. Shared conceptualisations
15. Pairing for requirements analysis
16. Retrospectives
17. Continuous planning

Example user stories in brief

- #1. As Admin, I'm able to import Event Records,
- #8. As a Visitor, I'm able to navigate the site through the site wide footer menu, So that I can always quickly open the page I'm looking for
- #9. As a Visitor, I'm able to navigate the site through the site wide top menu, So that I can always quickly open the page I'm looking for
- #10. As a Visitor, I'm able to navigate the site through site wide navigation menus, So that I can always quickly open the page I'm looking for
- #14. As a Visitor, I am able to use the contact form, So that I can contact the administrator
- #16. As an Administrator, I'm able to ban a particular User, So that he/she has no longer access to the site with the provided email address
- #17. As an Administrator, I'm able to completely remove a User from the site, So that the account is no longer available
- #18. As an Administrator, I can edit the details of a User,
- #20. As an Administrator, I'm able to search through the list of Users, So that I can more easily find a particular User
- #21. As an Administrator, I'm able to see a list of active Users registered with the site, So that I can manage the Users
- #23. As an Administrator, I'm able to manage Users,
- #24. As a User, I am able to set a new password, So that I can login

Summary

- ◆ RE involves elicitation, specification, validation, and negotiation
- ◆ Elicitation has five iterative steps, four of which are preparation steps. We use multiple elicitation techniques.
- ◆ We not only collect the requirements but also shape them
 - Analyzing requirements by structuring them
 - Negotiating with stakeholders in requirement elicitation and prioritization
 - Validating requirements from multiple sources

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